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Fault tolerant attitude control of under-actuated spacecraft: Theory and experiment

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Abstract This paper investigates fault tolerant attitude control theory and experiment for under-actuated spacecraft with one reaction wheel completely broken and two others suffering actuator faults of partial loss of effectiveness or bias. A non-smooth robust adaptive fault tolerant control law is proposed under the zero-momentum and input saturation conditions. It shows that the available reaction wheels need to produce sufficient control torque for the fault tolerance. Such a new control method is implemented in a semi-physical simulation system of an air-bearing platform. Experimental results show the effectiveness of the proposed method in spacecraft practical engineering.

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1. Introduction

Spacecraft attitude control is an important issue to realize advanced space missions such as earth gravity measurement, space observation, and space scientific experiments. During past years, there has been a large amount of research work on spacecraft attitude control that guarantees rapid, accurate, and global response according to different space missions^{1–3}.

Various faults often occur inevitably in spacecraft attitude systems due to working in the harsh space environments. Actuator fault, which represents one of the most serious and common threats to spacecraft operations, not only results in

performance degradation of spacecraft, but also leads to system instability. During past decades, there has been various work devoted to Fault Tolerant Control (FTC) algorithm design for spacecraft under actuator faults, such as backstepping control,⁴ sliding-mode control,⁵ time-delay control methods,⁶ robust control designs,^{7–8} adaptive-based designs,^{9–11} fixed-time and finite-time FTC methods.^{12–16} Readers can refer to a review¹⁷ for more details. However, the existing methods often rely on the redundant reaction wheel configuration, which makes fault compensation and control allocation more convenient. This, however, requires enough available hardware even in the faulty situations, and thus does not work in the absence of redundant actuators.

Particularly, the total loss of effectiveness in certain spacecraft actuators will transform the fully actuated spacecraft into an under-actuated one, whose control problem is more complicated and challenging because such under-actuated systems are nonholonomic and cannot be asymptotically stabilized using smooth time-invariant control laws.¹⁸ There has been some work about under-actuated spacecraft attitude control. Refs.

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^{19, 20} considered the attitude control of a satellite with two reaction wheels. Both of them considered under-actuated spacecraft in a zero-momentum condition,²¹ in which three-axis stabilization was shown to be feasible by two control torques. For a more general condition on the spacecraft dynamic model, Ref.²² proposed a new algorithm for under-actuated spacecraft attitude control based on tube-based model predictive control. Ref.²³ studied the model predictive control stabilization problem without considering external disturbances in practical engineering. Ref.²⁴ proposed a discontinuous distributed control algorithm for multiple under-actuated rigid spacecraft while Ref.²⁵ addressed the leader–follower tracking problem for a team of under-actuated flexible spacecraft by treating them as an under-actuated Lagrange system. However, these results do not consider the case that two available reaction wheels are also in the faulty situations of partial loss of effectiveness or bias. In this challenging case, the main difficulty lies in the non-smooth FTC strategy design to compensate for fault effects and realize three-axis stabilization using only two faulty reaction wheels.

On the other hand, inevitable constraints that are often neglected in theoretical analysis exist in practical systems such as low sampling time, data transmission delay and sensor bias. It is important and necessary to further evaluate attitude control schemes in practical engineering. A review of air-bearing spacecraft simulators was given in Ref.²⁶ Ref.²⁷ used an air-bearing spacecraft simulator to investigate the attitude control of spacecraft with sliding mode observers to investigate angular velocities. Ref.²⁸ used a satellite simulator to evaluate the performance of a finite time passivity based adaptive sliding mode control law. These experiments were all conducted to test algorithms in fault-free situations. Without considering the abrupt fault occurrence that may bring unpredictable effects to the platform, Ref.²⁹ validated fault tolerant attitude control methods via a reaction wheel assembly while the spacecraft model was still simulated in Matlab. Ref.³⁰ used a semi-physical three-axis experimental platform to verify the effectiveness of an observer-based FTC scheme for spacecraft subjected to actuator faults. However, these experiments focus on the system equipped with four reaction wheels for the purpose of fault tolerance.

Based on the above analysis, this paper mainly focuses on FTC theory and experiment of under-actuated spacecraft with one reaction wheel completely broken and two others suffering partial loss of effectiveness or bias. To the best of our knowledge, no result has been reported along this direction prior to this work.

The main contributions of this paper are summarized in two aspects:

- (1) A non-smooth robust adaptive FTC algorithm is proposed under the zero-momentum condition to stabilize three-axis attitude using only two faulty reaction wheels. A fault tolerance condition on maximum control input, fault and virtual control input is established. The presented algorithm treats spacecraft dynamic and kinematic models as a cascade system where the desired angular velocities are tracked by a developed non-smooth control law, and thus satisfies Brockett's condition and does not require fairly complex fault detection or isolation scheme. The new result shows that the avail-

able reaction wheels need to produce sufficient control torque for fault tolerance.

- (2) The proposed FTC algorithms are successfully implemented in the spacecraft semi-physical simulation system of an air-bearing platform with the consideration of actuator fault, modeling error, data transmission delay and parameter measurement bias. Particularly, zero-momentum condition for under-actuated spacecraft is simulated by providing the reaction wheels with extra control inputs to compensate for momentum. The experimental results verify the proposed methods, which is of great significance in spacecraft practical engineering.

The rest of this paper is organized as follows: [Section 2](#) introduces the spacecraft semi-physical simulation system and spacecraft model. [Section 3](#) develops the FTC theory. [Sections 4](#) gives numerical and experimental simulation results, followed by a conclusion in [Section 5](#).

2. Simulation system and spacecraft model

This section introduces the spacecraft semi-physical simulation system and spacecraft kinematic and dynamic models with actuator faults. The spacecraft semi-physical simulation system can simulate three-axis motion of rigid spacecraft to confirm control algorithms under a more realistic condition in space.

2.1. Spacecraft semi-physical simulation system

[Fig. 1](#) shows the whole picture of the spacecraft semi-physical simulation system, which consists of 4 parts: the air-bearing platform, the air source, the vision measurement system, and a ground station.



Fig. 1 Overview of spacecraft semi-physical simulation system.

The overview of the air-bearing platform is shown in Fig. 2, which is composed of the following:

- (1) On-board computer, receiving desired attitude and control instructions from the ground computer and implementing actual control instructions.
- (2) Actuators, including reaction wheels and gas propulsion system.
- (3) Sensors, including a gyro and an inertial sensor to measure angular velocity and attitude respectively.
- (4) Automatic balance regulation system, compensating for disturbance moment and balancing the platform.

Fig. 3 shows the configuration of Reaction Wheels (RWs). Four reaction wheels are installed in a traditional way. Three orthogonal reaction wheels labeled by RW1, RW2 and RW3 can control three-axis motion of the air-bearing platform. The oblique reaction wheel labeled by RW4 is redundant in the equiangular direction. In this paper, we only use RW1, RW2 and RW3 as actuators without actuating RW4.

Fig. 4 briefly summarizes the overall operation process of this spacecraft semi-physical simulation system. The ground station sends target attitude and control instructions to the on-board computer in the air-bearing platform through wireless connection. Then the on-board computer implements maneuvering commands to reaction wheels to actuate the platform. The gyro and inertial sensors measure the attitude of the platform and send state information in the form of Euler angles back to the on-board computer. By comparing sensor outputs with the target attitude, the on-board computer calculates the control torque commands and applies them to the active actuators.

2.2. Spacecraft kinematics and dynamics

The attitude kinematics and dynamics of the air-bearing platform, which simulate a rigid spacecraft, are represented by the unit quaternion with

$$\begin{cases} \dot{q}_0 = -\frac{1}{2} \mathbf{q}^T \boldsymbol{\omega} \\ \dot{\mathbf{q}} = \frac{1}{2} (\mathbf{q}^\times + q_0 \mathbf{I}_3) \boldsymbol{\omega} \\ \mathbf{J} \dot{\boldsymbol{\omega}} = -\boldsymbol{\omega}^\times \mathbf{J} \boldsymbol{\omega} + \mathbf{D} \mathbf{u} + \mathbf{d} \end{cases} \quad (1)$$

where $\boldsymbol{\omega} \triangleq [\omega_1, \omega_2, \omega_3]^T \in \mathbb{R}^3$ denotes the bounded spacecraft angular velocity vector with respect to an inertial frame $\mathcal{I}$,

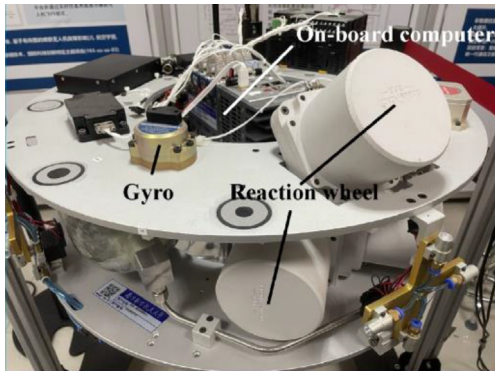


Fig. 2 Overview of air-bearing platform.

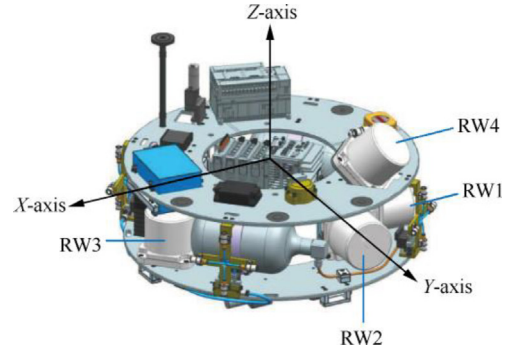


Fig. 3 Configuration of four Reaction Wheels (RWs).

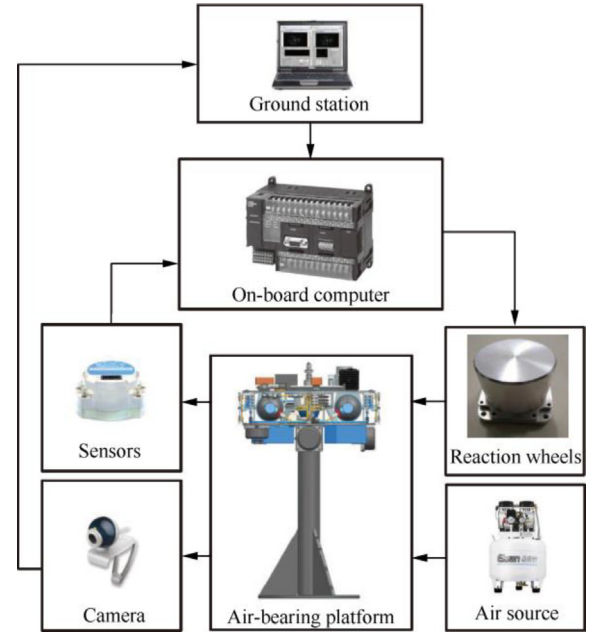


Fig. 4 Illustration of control loop.

$\bar{\mathbf{q}} \triangleq [q_0, \mathbf{q}^T]^T \in \mathbb{R}^4$ with $\mathbf{q} \triangleq [q_1, q_2, q_3]^T \in \mathbb{R}^3$ denotes attitude orientation of spacecraft in the body-fixed frame $\mathcal{B}$ with respect to the inertial frame. q_0 and $\mathbf{q}$ satisfy $q_0^2 + \mathbf{q}^T \mathbf{q} = 1$, $\mathbf{I}_3$ denotes the 3×3 identity matrix. $\mathbf{J} \in \mathbb{R}^{3 \times 3}$ represents the positive-definite inertial matrix of the spacecraft, $\mathbf{u} \triangleq [u_1, u_2, u_3, u_4]^T \in \mathbb{R}^4$ denotes the control inputs generated by reaction wheels, $\mathbf{d} \triangleq [d_1, d_2, d_3, d_4]^T \in \mathbb{R}^4$ denotes the external disturbance, $\mathbf{D} \in \mathbb{R}^{3 \times 4}$ denotes the torque distribution matrix with

$$\mathbf{D} = \begin{bmatrix} 1 & 0 & 0 & 1/\sqrt{3} \\ 0 & 1 & 0 & 1/\sqrt{3} \\ 0 & 0 & 1 & 1/\sqrt{3} \end{bmatrix} \quad (2)$$

Since only RW1, RW2 and RW3 are used, $u_4 = d_4 = 0$. The symbol $\mathbf{a}^\times$ with $\mathbf{a} = [a_1, a_2, a_3]^T$ denotes a skew-symmetric matrix with

$$\mathbf{a}^\times = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \quad (3)$$

2.3. Actuator fault model

Consider actuator faults of both loss of effectiveness and bias in reaction wheels on X , Y , and Z axes that are modeled as

$$\mathbf{u} = \sigma \mathbf{v} + \phi \quad (4)$$

where $\mathbf{v} \triangleq [v_1, v_2, v_3, 0]^T \in \mathbb{R}^4$ is the FTC strategy to be designed. $\sigma \triangleq \text{diag}(\sigma_1, \sigma_2, \sigma_3, 0) \in \mathbb{R}^{4 \times 4}$ with $0 \leq \sigma_i \leq 1$ denotes the loss of effectiveness in the i th reaction wheel for $i = 1, 2, 3$. Note that $\sigma_i = 1$ means that the i th reaction wheel works normally; $0 < \sigma_i < 1$ means that the i th reaction wheel partially loses its effectiveness but still works; $\sigma_i = 0$ means that the i th reaction wheel is completely broken and the spacecraft becomes under-actuated. $\phi \triangleq [\phi_1, \phi_2, \phi_3, 0]^T \in \mathbb{R}^4$ denotes the bias fault. In this paper, it is assumed that the reaction wheel labeled by RW3 is completely broken, which means $\sigma_3 = \phi_3 = 0$. Considering the experimental conditions in practice, the control inputs generated by the reaction wheels are limited by $|v_i| \leq v_{\max}$ for $i = 1, 2$. It is reasonable to assume that the actuator faults are both bounded, i.e., $|\phi_i(t)| \leq \bar{\phi}_i$ for $i = 1, 2$, and there exists a constant $\lambda_{\min}$ satisfying $0 \leq \lambda_{\min} \leq \min_{i=1,2} \{\sigma_i\}$.

2.4. Zero-momentum condition

In this paper, the spacecraft is considered to be under-actuated in the beginning and operate in a zero-momentum condition.^{19,20} This means that no external force has been applied to spacecraft and the disturbance torque is assumed to be zero, i.e., $d_i = 0$ for $i = 1, 2, 3, 4$, and thus the momentum of spacecraft is conserved with zero initial value. Moreover, when a symmetric spacecraft, which has a symmetric mass distribution about their certain axis, encounters actuator failures and loses the control torque along the symmetry axis, the spacecraft will enter this mode and remain rotating about the symmetry axis at a constant rate.³¹ Related researches for under-actuated spacecraft in this condition have been conducted and tested in in-orbit experiments on spacecraft Uosat-12 and UK-DMC1^{13,14}

As introduced in Ref.,¹⁹ for under-actuated spacecraft under a zero-momentum condition, the reaction wheels are assumed to be assembled in the spacecraft principal axes. Under this precondition, the inertial matrix becomes $\mathbf{J} = \text{diag}(J_1, J_2, J_3)$ with $J_i > 0$. Zero-total angular momentum mode is considered, which means that in the dynamical model Eq. (1), one has

$$\omega^\times \mathbf{J} \omega = \mathbf{0} \quad (5)$$

On the unactuated axis, the aforementioned condition also implies $\omega_3 = 0$.

Remark 1. In our air-bearing platform, three orthogonal reaction wheels are not assembled in spacecraft principal axes as shown in Fig. 3. Thus, the inertial matrix $\mathbf{J}$ of the actual platform is only a general skew-symmetric matrix and not a

diagonal matrix. Moreover, $\omega^\times \mathbf{J} \omega = \mathbf{0}$ is not satisfied since this condition is difficult to strictly achieve on earth. However, if we give reaction wheels extra control commands to compensate these terms as.

$$\begin{cases} \tau_1 = J_{12}\dot{\omega}_2 + J_{13}\dot{\omega}_3 + (\omega^\times \mathbf{J} \omega)_1 \\ \tau_2 = J_{21}\dot{\omega}_1 + J_{23}\dot{\omega}_3 + (\omega^\times \mathbf{J} \omega)_2 \\ \tau_3 = J_{31}\dot{\omega}_1 + J_{32}\dot{\omega}_2 + (\omega^\times \mathbf{J} \omega)_3 \end{cases} \quad (6)$$

where J_{ij} denotes the element of the i th line and the j th column in inertial matrix $\mathbf{J}$ and $(\omega^\times \mathbf{J} \omega)_i$ denotes the i th line of the column $\omega^\times \mathbf{J} \omega$ for $i, j = 1, 2, 3$, then the platform can approximately realize this zero-momentum condition with $J_i = J_{ii}$ for $i = 1, 2, 3$ and FTC of under-actuated spacecraft under such a condition can be experimented.

Thus, by applying zero-momentum condition Eq. (5) and $\omega_3 = 0$ to the aforementioned rigid spacecraft model Eq. (1) under actuator faults Eq. (4) where $\sigma_3 = \phi_3 = 0$, the dynamics and kinematics for under-actuated spacecraft can be rewritten as

$$\begin{cases} J_1\dot{\omega}_1 = \sigma_1 v_1 + \phi_1 \\ J_2\dot{\omega}_2 = \sigma_2 v_2 + \phi_2 \\ J_3\dot{\omega}_3 = 0 \end{cases} \quad (7)$$

and

$$\begin{cases} \dot{q}_0 = -\frac{1}{2}q_1\omega_1 - \frac{1}{2}q_2\omega_2 \\ \dot{q}_1 = \frac{1}{2}q_0\omega_1 - \frac{1}{2}q_3\omega_2 \\ \dot{q}_2 = \frac{1}{2}q_3\omega_1 + \frac{1}{2}q_0\omega_2 \\ \dot{q}_3 = -\frac{1}{2}q_2\omega_1 + \frac{1}{2}q_1\omega_2 \end{cases} \quad (8)$$

where ω_1 and ω_2 can be treated as the virtual control inputs for kinematic model Eq. (8).

Control objective: Considering under-actuated spacecraft Eqs. (7) and (8) when reaction wheel RW3 is completely broken and the other two still suffer from actuator faults, we design a FTC strategy v_i for $i = 1, 2$ so that the under-actuated spacecraft can realize stabilization in a fixed time T , i.e.,

$$\begin{cases} \lim_{t \rightarrow \infty} q_0(t) = 1 \\ \lim_{t \rightarrow \infty} \mathbf{q}(t) = \mathbf{0} \\ \lim_{t \rightarrow \infty} \omega(t) = \mathbf{0} \end{cases} \quad (9)$$

3. FTC algorithm design

This section addresses FTC design issue to achieve the control objective Eq. (9). The under-actuated spacecraft model is treated as a cascade system and a non-smooth robust adaptive FTC law is proposed under the zero-momentum and input saturation conditions.

To stabilize the kinematic model Eq. (8), a virtual controller in a fault-free situation is proposed in Ref.³² as

$$\begin{cases} \omega_{d1} = -aq_1 + b\frac{q_2q_3}{q_1+q_2} \\ \omega_{d2} = -aq_2 - b\frac{q_1q_3}{q_1+q_2} \end{cases} \quad (10)$$

where a and b are positive constants. It has been proved in Ref.¹⁹ that singularity in the control law Eq. (10) can be ruled out by setting $b > 4a$. Such a virtual control law can be

regarded as desired angular velocities of under-actuated spacecraft. $\omega_{d3} = 0$ holds in the under-actuated axis. Note that $\dot{\omega}_{di}$ is bounded and will be proved in the following. Define L_i as the bound of $\dot{\omega}_{di}$ for $i = 1, 2$.

For the under-actuated spacecraft under actuator faults, the following theorem provides a robust adaptive fault tolerant controller, and establishes a fault tolerance condition on bias fault, loss of effectiveness, desired virtual control torque and control input bounds.

Theorem 1. Consider the under-actuated spacecraft in a zero-momentum condition with its dynamic and kinematic models given in Eqs. (7) and (8). Design the robust adaptive FTC law as.

$$v_i = -\frac{v_{\max}(\omega_i - \omega_{di})}{|\omega_i - \omega_{di}| + k^2\delta} \quad i = 1, 2 \quad (11)$$

and the adaptive law $k > 0$ as

$$\dot{k} = \frac{\gamma k}{1 + 4\gamma(1 - q_0)} \cdot \left[-\sum_{i=1}^2 |\omega_i - \omega_{di}| \frac{\lambda_{\min} v_{\max}(\delta + 1)}{|\omega_i - \omega_{di}| + k^2(\delta + 1)} \right] \quad (12)$$

where $\delta > 0$ is a small positive constant and $\gamma > 0$ is a designed positive constant. If the bounds $\bar{\phi}_i$, L_i , $\lambda_{\min}$ and the maximum control input $v_{\max}$ satisfy

$$\bar{\phi}_i + J_i L_i \leq \lambda_{\min} v_{\max} \quad i = 1, 2 \quad (13)$$

then $\lim_{t \rightarrow \infty} q_0(t) = 1$, $\lim_{t \rightarrow \infty} q(t) = \mathbf{0}$ and $\lim_{t \rightarrow \infty} \omega(t) = \mathbf{0}$.

Before proving Theorem 1, we provide some insight in controller Eq. (11) and condition Eq. (13).

Remark 2. The design idea of this FTC law (11) comes from the sliding mode control law $v_i = -v_{\max} s_i / (|s_i| + \delta)$ for spacecraft attitude control in Ref.³³ The actuator fault effects are eliminated by parameter adaption of k . By setting $k(0) > 0$ and deriving k from term cancellation in stability analysis, it can be proved in the following that k is always greater than a positive constant. From the expression of the FTC law, it is obvious that the proposed FTC law satisfies reaction wheel torque limits, i. e., $|v_i| \leq v_{\max}$, for $i = 1, 2$, which is practically meaningful.

Remark 3. The physical meaning of $J_i \dot{\omega}_{di}$ is the desired virtual control torque to stabilize the kinematic model. Thus, in condition Eq. (13), $J_i L_i$ for $i = 1, 2$ denotes the bound of desired virtual control torque needed to stabilize two actuated axes. The physical meaning of condition Eq. (13) is that the reaction wheels under loss of effectiveness are still capable of supplying sufficient control torque to reject bias faults and realize kinematic stabilization under the virtual controller Eq. (10). Otherwise, the bias faults cannot be compensated.

Proof. Consider the Lyapunov candidate as

$$\begin{aligned} V &= \frac{1}{2}(\omega - \omega_d)^T J(\omega - \omega_d) \\ &+ k^2 \left[q^T q + (1 - q_0)^2 \right] + \frac{1}{2\gamma} k^2 \\ &= \frac{1}{2}(\omega^T J(\omega - \omega_d) \\ &+ 2k^2 k^2(1 - q_0) + \frac{1}{2\gamma} k^2 \end{aligned} \quad (14)$$

Note that under the zero-momentum condition, the third terms of the vectors ω and ω_d are zero with $\omega = [\omega_1, \omega_2, 0]^T$ and $\omega_d = [\omega_{d1}, \omega_{d2}, 0]^T$.

The time derivative of V along the solution of Eqs. (7) and (8) is

$$\begin{aligned} \dot{V} &= (\dot{\omega} - \dot{\omega}_d)^T J(\dot{\omega} - \dot{\omega}_d) \\ &- 2k^2 \dot{q}_0 + 4k\dot{k}(1 - q_0) + \frac{1}{\gamma} k\dot{k} \end{aligned} \quad (15)$$

To cancel $\dot{q}_0$ in Eq. (15), substituting the virtual controller Eq. (10) into Eq. (8), one has:

$$\begin{aligned} \dot{q}_0 &= -\frac{1}{2} q_1 \left(-aq_1 + b \frac{q_2 q_3}{q_1^2 + q_2^2} \right) \\ &- \frac{1}{2} q_2 \left(-aq_2 - b \frac{q_1 q_3}{q_1^2 + q_2^2} \right) \\ &= \frac{1}{2} a(q_1^2 + q_2^2) \geq 0 \end{aligned} \quad (16)$$

Thus, substituting Eq. (7) and the FTC law Eq. (11) into Eq. (15) yields:

$$\begin{aligned} \dot{V} &= (\dot{\omega} - \dot{\omega}_d)^T [(\sigma v + \phi) - J\dot{\omega}_d] \\ &- k^2 a(q_1^2 + q_2^2) + k\dot{k}[4(1 - q_0) + \frac{1}{\gamma}] \end{aligned} \quad (17)$$

Since ω_d and $\dot{\omega}_d$ arise in Eq. (17), we first prove the boundedness of them. For ω_{d1} , it is obvious that

$$\omega_{d1} \leq a|q_1| + b \left| \frac{q_2 q_3}{q_1^2 + q_2^2} \right| \quad (18)$$

From the analysis of the singularity for the controller Eq. (10) in Ref.,¹⁹ it has been proved that as the system approaches the origin, one has:

$$\lim_{t \rightarrow \infty} \frac{q_3(t)}{q_1^2(t) + q_2^2(t)} = \lim_{t \rightarrow \infty} \frac{q_3(0)e^{-\frac{bt}{2}}}{c_1 e^{-at} + c_2 e^{-bt}} \quad (19)$$

where c_1 and c_2 are certain constants. The numerator of the right-hand side of Eq. (19) converges faster to 0 than the denominator if $b > 2a$. Since $q_i \leq 1$ for $i = 1, 2, 3, 4$, it can be concluded that ω_{d1} is bounded. Similar conclusion can also be obtained for ω_{d2} .

To prove the boundedness of $\dot{\omega}_{di}$ in Eq. (17), taking the time derivative of ω_{d1} along the solution of Eq. (8) yields:

$$\begin{aligned} \dot{\omega}_{d1} &= b \frac{(\dot{q}_2 q_3 + \dot{q}_3 q_2)(q_1^2 + q_2^2) - (2q_1 \dot{q}_1 + 2q_2 \dot{q}_2)q_2 q_3}{(q_1^2 + q_2^2)^2} \\ &- a\dot{q}_1 \end{aligned} \quad (20)$$

Substituting Eq. (10) into Eq. (8), we have

$$\dot{q}_3 = -\frac{b}{2} q_3 \quad (21)$$

So we can obtain that $q_3(t) = q_3(0)e^{-bt/2}$. Substituting Eq. (21) into Eq. (20), one has

$$\begin{aligned} \dot{\omega}_{d1} &= -a\dot{q}_1 + \frac{bq_3}{(q_1^2 + q_2^2)^2} (\dot{q}_2 - 0.5bq_2)(q_1^2 + q_2^2) \\ &- \frac{bq_2 q_3}{(q_1^2 + q_2^2)^2} (2q_1 \dot{q}_1 + 2q_2 \dot{q}_2) \end{aligned} \quad (22)$$

Since q_1, q_2 , ω_{d1} and ω_{d2} are bounded, from Eq. (8), it is obvious that $\dot{q}_1$ and $\dot{q}_2$ are also bounded.

Thus, $(\dot{q}_2 - 0.5bq_2)(q_1^2 + q_2^2) - q_2(2q_1 \dot{q}_1 + 2q_2 \dot{q}_2)$ is bounded. Considering the singular term in Eq. (10) alone, as the system approaches the origin, one has:

$$\begin{aligned} &\lim_{t \rightarrow \infty} \frac{q_3(t)}{[q_1^2(t) + q_2^2(t)]^2} \\ &= \lim_{t \rightarrow \infty} \frac{q_3(0)e^{-\frac{bt}{2}}}{c_1^2 e^{-2at} + c_2^2 e^{-2bt} + 2c_1 c_2 e^{-(a+b)t}} \end{aligned} \quad (23)$$

If we set $b > 4a$, then the numerator of the right-hand side of Eq. (23) converges to 0 faster than the denominator. Therefore, it leads to:

$$\lim_{t \rightarrow \infty} \frac{q_3(t)}{[q_1^2(t) + q_2^2(t)]^2} = 0 \quad (24)$$

Then, $\dot{\omega}_{d1}$ is bounded. Similar conclusion can be obtained for $\dot{\omega}_{d2}$.

Note that the following inequality holds:

$$\begin{aligned} & -\sigma_i \cdot \frac{v_{\max}(\omega_i - \omega_{di})^2}{|\omega_i - \omega_{di}| + k^2 \delta} \\ & \leq -\lambda_{\min} v_{\max} \cdot \frac{(\omega_i - \omega_{di})^2}{|\omega_i - \omega_{di}| + k^2(\delta + 1)} \\ & \leq -\lambda_{\min} v_{\max} |\omega_i - \omega_{di}| \left[1 - \frac{k^2(\delta + 1)}{|\omega_i - \omega_{di}| + k^2(\delta + 1)} \right] \end{aligned} \quad (25)$$

Substituting inequality Eq. (25) and L_i into Eq. (17) yields:

$$\begin{aligned} \dot{V} &= \sum_{i=1}^2 (\omega_i - \omega_{di}) [(\sigma_i v_i + \phi_i) - J_i \dot{\omega}_{di}] \\ &\quad - k^2 a (q_1^2 + q_2^2) + k \dot{k} [4(1 - q_0) + \frac{1}{\gamma}] \\ &\leq -\sum_{i=1}^2 |\omega_i - \omega_{di}| (\lambda_{\min} v_{\max} - \bar{\phi}_i - J_i L_i) \\ &\quad - k^2 a (q_1^2 + q_2^2) + k \dot{k} [4(1 - q_0) + \frac{1}{\gamma}] \\ &\quad + \sum_{i=1}^2 |\omega_i - \omega_{di}| \frac{\lambda_{\min} v_{\max} k^2 (\delta + 1)}{|\omega_i - \omega_{di}| + k^2 (\delta + 1)} \end{aligned} \quad (26)$$

Substituting the updating law Eq. (12), it follows that

$$\dot{V} \leq -\sum_{i=1}^2 |\omega_i - \omega_{di}| (\lambda_{\min} v_{\max} - \bar{\phi}_i - J_i L_i) - k^2 a (q_1^2 + q_2^2) \quad (27)$$

Condition Eq. (13) guarantees that $\dot{V} \leq 0$. This also implies that $\omega - \omega_d$ and k are bounded. Integrating both sides of inequality Eq. (27) leads to:

$$\begin{aligned} V(\infty) - V(0) &\leq -c_3 \int_0^\infty \|\omega(t) - \omega_d(t)\| dt \\ &\quad - c_3 a \int_0^\infty k^2(t) (q_1^2(t) + q_2^2(t)) dt \end{aligned} \quad (28)$$

where $c_3 \triangleq \min\{\lambda_{\min} v_{\max} - \bar{\phi}_i - J_i L_i, 1\}$ and $\|\cdot\|$ denote the 1-norm of vectors. Because the term on the left-hand side of inequality Eq. (28) is bounded, it follows that $\omega \in \mathcal{L}_1$ and $k\sqrt{q_1^2 + q_2^2} \in \mathcal{L}_2$, where $\mathcal{L}_1$ and $\mathcal{L}_2$ denote $\mathcal{L}_1$ -norm and $\mathcal{L}_2$ -norm, respectively. Note that $\dot{k}$, $\dot{\omega}$ and $\dot{q}$ are all bounded. Then using Barbalat's lemma, it follows that

$$\lim_{t \rightarrow \infty} (\omega(t) - \omega_d(t)) = \lim_{t \rightarrow \infty} k(t) \sqrt{q_1^2(t) + q_2^2(t)} = 0 \quad (29)$$

This result indicates that $\lim_{t \rightarrow \infty} q_1^2(t) + q_2^2(t) = 0$ if $\lim_{t \rightarrow \infty} k(t) \neq 0$.

Next we will discuss whether k can converge to 0.

From Eq. (12), one can obtain

$$\begin{aligned} \dot{k} &\geq -\gamma \lambda_{\min} v_{\max} (\delta + 1) k \sum_{i=1}^2 \frac{|\omega_i - \omega_{di}|}{|\omega_i - \omega_{di}| + k^2 (\delta + 1)} \\ &\geq -2\gamma \lambda_{\min} v_{\max} (\delta + 1) k \end{aligned} \quad (30)$$

Hence, it follows that

$$k(t) \geq k(0) e^{-\gamma c_4 t} \quad (31)$$

where $c_4 \triangleq 2\lambda_{\min} v_{\max} (\delta + 1) > 0$. Inequality Eq. (31) shows that k is bounded and always greater than 0. On the other hand, it holds that

$$\begin{aligned} \dot{k} &\geq -\gamma k \left[\sum_{i=1}^2 |\omega_i - \omega_{di}| \cdot \frac{\lambda_{\min} v_{\max} (\delta + 1)}{|\omega_i - \omega_{di}| + k^2 (\delta + 1)} \right] \\ &\geq -\gamma k \left[\sum_{i=1}^2 |\omega_i - \omega_{di}| \cdot \frac{\lambda_{\min} v_{\max} (\delta + 1)}{k^2 (\delta + 1)} \right] \\ &\geq -\frac{\gamma}{k} \left[\sum_{i=1}^2 |\omega_i - \omega_{di}| \lambda_{\min} v_{\max} \right] \end{aligned}$$

It follows that

$$k \dot{k} \geq -2\gamma \lambda_{\min} v_{\max} \|\omega - \omega_d\| \quad (33)$$

Integrating this inequality leads to

$$k^2(\infty) - k^2(0) \geq -4\gamma \lambda_{\min} v_{\max} \int_0^\infty \|\omega(t) - \omega_d(t)\| dt \quad (34)$$

Since $\omega - \omega_d$ is bounded, we can always find a γ so that there exists a positive function $f_1(\gamma)$ satisfying

$$\int_0^\infty \|\omega(t) - \omega_d(t)\| dt \leq f_1(\gamma) \quad (35)$$

Therefore, for any given $k(0) > 0$ and $\underline{k} > 0$ satisfying $k(0) > \underline{k}$, we can find a small γ such that

$$k^2(0) - \underline{k}^2 \geq 4\gamma \lambda_{\min} v_{\max} f_1(\gamma) \quad (36)$$

Thus, $k > \underline{k} > 0$ can be obtained. Therefore, it can be concluded that $\lim_{t \rightarrow \infty} q_1^2(t) + q_2^2(t) = 0$. From Eq. (8), $\lim_{t \rightarrow \infty} q_3(t) = 0$ holds. Then based on $q_0^2 + q^T q = 1$, $\lim_{t \rightarrow \infty} q_0(t) = 1$ can be obtained. The control objective Eq. (9) is achieved. This completes the proof.

Remark 4. In the above theorem, symbols c_1, c_2, c_3, c_4 and $\underline{k}$ are only used for stability analysis. a and b satisfying $b > 4a > 0$ are designed coefficients in the virtual controllers Eq. (10). From Eq. (22), it can be seen that the bound of $\dot{\omega}_{di}$ for $i = 1, 2$ is proportional to a and b . Thus, by setting a and b as small as possible, condition Eq. (13) can always be satisfied if $\lambda_{\min} v_{\max} > \bar{\phi}_i$. $\delta > 0$ is an arbitrary positive constant. γ also needs to be designed in the adaptive law Eq. (12). Inequality Eq. (35) imposes an implicit condition that γ needs to satisfy to guarantee $\gamma > \underline{\gamma} > 0$. Although giving an explicit function $f_1(\gamma)$ is preferable, it is difficult to find such a solution as shown in Ref.¹³ However, in the simulation, when γ is chosen as $\gamma \ll 1$, the existence of a positive lower bound of k can always be assured.

Remark 5. The proposed FTC law Eq. (11), the virtual control law Eq. (10) and the adaptive law Eq. (12) do not use the information of inertial matrix of the under-actuated spacecraft, and thus the proposed FTC law is robust to parameter uncertainties and these uncertainties will not cause the system performance to deteriorate. Moreover, the proposed FTC control strategy uses the sliding mode control structure and it is a discontinuous time-invariant control law, which satisfies Brockett's condition.

4. Numerical simulation and experiment

In this section, a numerical simulation and a semi-physical experiment are conducted to verify the effectiveness of the proposed FTC scheme.

Table 1 Parameters of air-bearing platform and reaction wheels.

Parameter	Value
Inertial matrix $J(\text{kg} \cdot \text{m}^2)$	$\begin{bmatrix} 0.878 & 0.044 & 0.023 \\ -0.044 & 0.962 & 0.032 \\ -0.023 & -0.032 & 1.474 \end{bmatrix}$
Sampling time (s)	0.2
Rotation speed limit (r/min)	± 4500
Control torque limit $\tau_{\max}(\text{N} \cdot \text{m})$	0.015

4.1. Numerical simulation

In this numerical simulation, the under-actuated spacecraft is allowed to operate in a perfect zero-momentum condition. The spacecraft parameters are identical to the air-bearing platform shown in Table 1. As illustrated in Remark 1, it follows that $J_1 = 0.878$, $J_2 = 0.962$ and $J_3 = 1.474$, which come from the diagonal elements of the platform inertial matrix. Table 2 gives the other control parameters in the simulation. Before giving simulation results, we make an explanation on parameter setting.

Remark 6. In order to compare control performance, parameters in the simulation and experiment are identical. However, due to the maximum control torque limitation of reaction wheels in our air-bearing platform, the experiment cannot run in a long time. Otherwise, the reaction wheels will exceed their maximum rotation speed limitation. Therefore, the fault occurrence time is set at an early time with $t = 25$ s and the initial state values are close to the origin.

For under-actuated spacecraft Eqs. (7) and (8), under above parameters and the FTC law Eq. (11), simulation results are clearly shown in Figs. 5-8. Figs. 5-7 show trajectories of spacecraft states and control inputs, from which we can see that the under-actuated spacecraft can realize three-axis stabilization using two reaction wheels in spite of faults. Fault effects on angular velocity ω at $t = 25$ s are magnified in

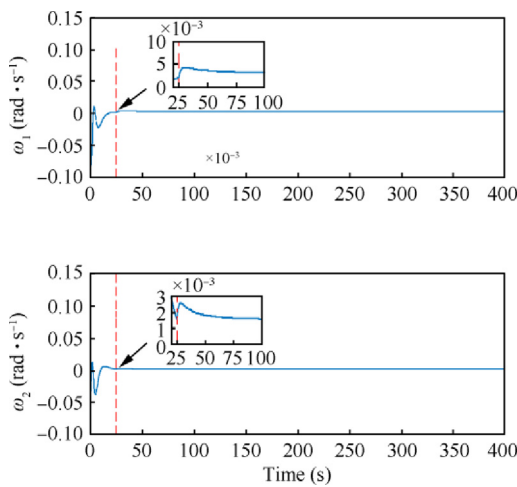
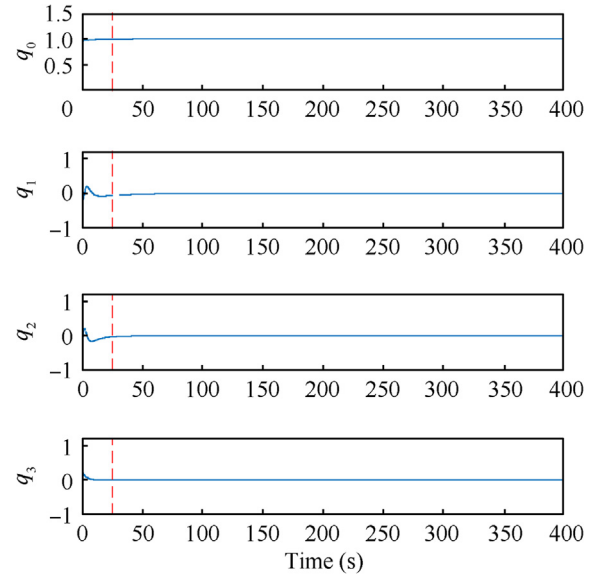
**Fig. 5** Simulation trajectories of angular velocity in actuated axes.**Fig. 6** Simulation trajectories of attitude $\bar{q}$.

Fig. 5. It can be seen that it takes about 50 s to stabilize the under-actuated spacecraft after fault occurrence at $t = 25$ s. Fig. 8 shows that the adaptive law $k(t)$ converges to a constant value greater than 0, which is consistent with its behavior analysis in Theorem 1. By checking workspace in Matlab, the bound of the virtual controller Eq. (10) is about $L_1 = 0.0069$ and $L_2 = 0.0043$, which satisfies the fault tolerance condition Eq. (13).

To the best of the authors' knowledge, there is no relevant result about fault tolerant control of such under-actuated spacecraft. Traditional attitude controller for under-actuated spacecraft in the fault-free situation cannot address actuator faults in the two actuated reaction wheels. Therefore, we do not give a further comparison with other approaches in simulation.

Overall, when the under-actuated spacecraft is in a perfect zero-momentum condition, numerical simulation results show an excellent control performance and robustness against faults.

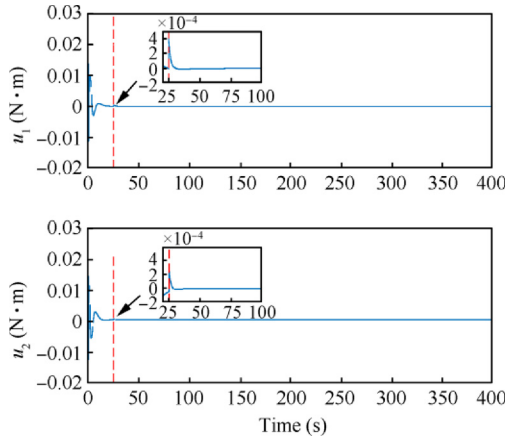
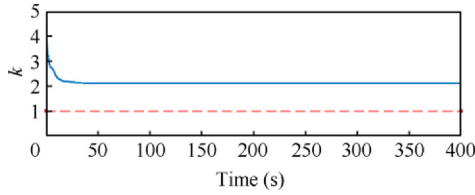
4.2. Experimental validation

Although the numerical simulation has validated the proposed FTC algorithm, inevitable constraints exist in physical spacecraft systems, such as parameter measurement bias, modeling error, disturbance torque and data transmission delay. To further verify the practical feasibility and effectiveness of the proposed FTC law, an experiment is conducted in the spacecraft semi-physical simulation systems in Fig. 1.

The proposed FTC algorithm Eq. (11) is embedded by Lab-View software in the ground station and is sent to the on-board computer in the platform through wireless connection. Before running the experiment, the platform is balanced using automatic balance regulation system. Since this platform is air-bearing, even if it is balanced well, the external disturbance torque cannot be completely compensated for and the platform cannot remain strictly static in practice. The automatic balance regulation system can only guarantee that the platform is in a

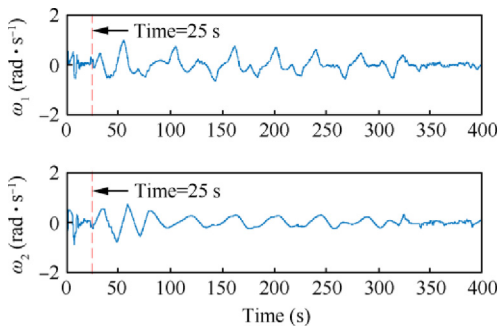
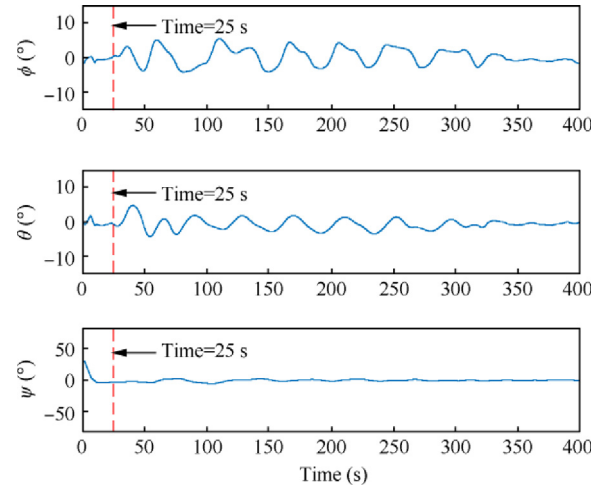
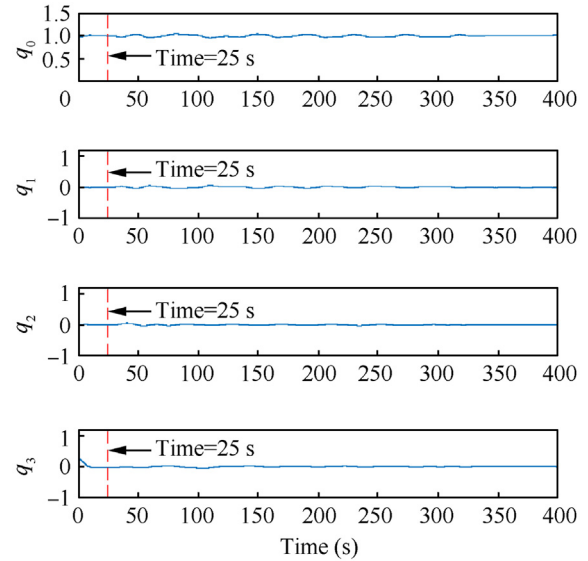
Table 2 Control parameters in numerical simulation.

Parameter	Value
Loss of effectiveness	$\sigma_1 = \begin{cases} 1 & t < 25 \\ 0.8 & t \geq 25 \end{cases}, \sigma_2 = \begin{cases} 1 & t < 25 \\ 0.9 & t \geq 25 \end{cases}$
Bias faults	$\phi_1 = \begin{cases} 0 & t < 25 \\ 0.0008 & t \geq 25 \end{cases}, \phi_2 = \begin{cases} 0 & t < 25 \\ 0.0007 & t \geq 25 \end{cases}$
Coefficients	$k(0) = 4, \delta = 0.2, \gamma = 0.001, a = 0.002, b = 0.009$
Initial states	$q_0(0) = 0.9671, q_1(0) = 0.0138, q_2(0) = 0.0055, q_3(0) = 0.2539, \omega_3(0) = 0, \omega_1(0) = -0.0669, \omega_2(0) = -0.0236$

**Fig. 7** Simulation trajectories of control input u .**Fig. 8** Simulation trajectories of adaptive law k .

simple pendulum motion with a time period about 50 s after balance.

Under the same control parameters and applying extra control commands as in Remark 1, experimental results are shown in Figs. 9–12. Figs. 9 and 10 show the trajectories of angular velocities and Euler angles of the air-bearing platform,

**Fig. 9** Experimental trajectories of angular velocity ω in actuated axes.**Fig. 10** Experimental trajectories of Euler angles ϕ, θ and ψ .**Fig. 11** Experimental trajectories of attitude $\bar{q}$ transformed from Euler angles.

where ϕ, θ and ψ denote roll, pitch and yaw angles respectively. These data are directly collected by the gyro and the inertial sensor. Fig. 11 gives trajectories of attitude $\bar{q}$ after transforming Euler angles into the unit quaternion. From these figures,

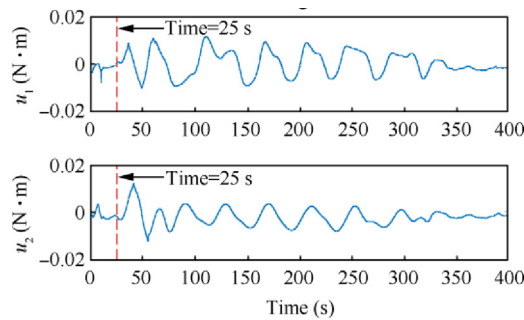


Fig. 12 Experimental trajectories of control input u .

we can see that the actuator faults have a greater influence on the practical spacecraft system and it takes about 300 s to stabilize the air-bearing platform.

Comparing experimental and numerical simulation results, we can see a longer settling time and more oscillations after fault occurrence in the experiment, which are caused by external disturbance torque and modeling error existing in the air-bearing platform. Moreover, data collection and transmission delay with sampling time 0.2 s also lead to inaccurate integration of the adaptive law $k(t)$, and thus the state information used in the FTC law Eq. (11) is not real-time and inaccurate. By contrast, in Fig. 7 of the numerical simulation, the control input u can perfectly track the virtual control input u in time, leading to less oscillations in state trajectories. Note that, in Fig. 11 of experimental results, $q_0 > 1$ happens occasionally, which contradicts with the definition of unit quaternion. This is caused by the attitude sensor measurement error in the practical air-bearing platform.

In summary, even there are inevitable constraints in practical spacecraft systems, experiments validate the effectiveness of the proposed FTC scheme for under-actuated spacecraft with acceptable performance. The proposed FTC scheme is practically useful and the engineering feasibility can be guaranteed.

5. Conclusions

In this paper, a robust adaptive fault tolerant controller has been provided for under-actuated spacecraft under the zero-momentum condition and a fault tolerance condition has been established on bias fault, loss of effectiveness, desired virtual control input and maximum control input. Numerical and experimental simulations have verified its effectiveness. However, the proposed robust fault tolerant control method can only address actuator faults with limited range depending on the maximum control inputs.

Future work will consider more practical active fault tolerant control laws for under-actuated spacecraft. Finite-time and fixed-time attitude control schemes with fast convergence rate will be researched. The consensus and tracking problems for multiple under-actuated spacecraft subjected to component faults and communication faults are also worth investigating.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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